Sparse-set extension and degree-sensitive perturbations for halves of triangle-free graphs

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Abstract

Erdős's sparse half conjecture asks whether every triangle-free graph on n vertices contains $\lfloor n/2 \rfloor$ vertices spanning at most $n^2/50$ edges. We give a computer-assisted proof of the general bound $(2543/100000)n^2 = 0.02543n^2$. The proof combines degree-sensitive neighbourhood perturbations, extension from a sparse vertex set, and neighbourhoods anchored across a maximum cut. A reciprocal-degree averaging criterion reduces one construction to an inequality in three real variables. Five exact rational subdivision certificates cover its entire parameter domain. The remaining ingredients are known maximum-cut bounds and a four-cycle density certificate, which we check with a separate implementation enumerating every triangle-free graph on six vertices.

Mathematics Subject Classifications: 05C35

1 Statement and relation to previous work

For a finite simple graph G on $n \geqslant 1$ vertices, put

$$b(G) = \min_{|S|=\lfloor n/2 \rfloor} e(G[S]), \qquad \rho(G) = \frac{2e(G)}{n^2}.$$

The sparse half conjecture asks whether $b(G) \leqslant n^2/50$ whenever G is triangle-free; see the problem entry [4]. Razborov [2] proved the general bound $27n^2/1024$. Sarid [3] subsequently gave a preprint claiming $131n^2/5000$, with a rational four-cycle certificate and a Lean development conditional on a quoted max-cut theorem. We use the numerical four-cycle certificate from that work, but neither import its verifier nor rely on its Lean development.

Theorem 1. *Every finite triangle-free graph G on $n \geqslant 1$ vertices satisfies*

$$b(G) \leqslant qn^2, \qquad q = \frac{2543}{100000} = 0.02543.$$

The numerical ingredients are a finite coefficient check and five interval checks. The first checks an existing four-cycle certificate; the others check a scalar inequality covering a continuum of parameters. None is a search restricted to graphs of bounded order. In particular, the earlier finite graph census in the accompanying research repository is not an assumption of Theorem 1.

The degree-dependent endpoint radii themselves already occur in [3]. Lemma 2 uses their dependence on the degree in a reciprocal-degree averaging criterion. Lemma 3 allows the starting set to contain edges. Section 5 uses neighbourhoods from the other side of the cut. We combine these with the density-specific max-cut bound. Neither the final constant nor the parameter 19/20 is claimed optimal.

2 Fractional halves and the external ingredients

Throughout the proof, G is finite, simple and triangle-free. Define

$$\beta_*(G) = \frac{1}{n^2} \min \left\{ \sum_{\{u,v\} \in E(G)} f_u f_v : 0 \leqslant f_v \leqslant 1, \quad \sum_v f_v = n/2 \right\}.$$

Then $b(G)/n^2 \leqslant \beta_*(G)$. Indeed, among minimizers choose one with as few fractional coordinates as possible. Moving two such coordinates as $(f_u + t, f_v - t)$ preserves their sum. The objective is affine or concave quadratic in t, so one endpoint of the feasible interval is no worse. Thus at most one coordinate is fractional. For even n all are integral; for odd n, discard the sole possible coordinate of value 1/2. We will use that $\beta_*(G(k)) = \beta_*(G)$ for every balanced blow-up $G(k)$. Indeed, replace a profile on each twin class by its average. The mass constraint scales by k and every edge product scales by k^2; conversely a profile on G lifts constantly to its twin classes.

Write $D_2(G)$ for the least number of edges lying within the two parts of a bipartition. We use the following consequences of Theorem 2(a,b) in the PDF version of [1]:

$$\frac{D_2(G)}{n^2} \leqslant \frac{2}{47}, \qquad \rho(G) \geqslant \rho_0 := \frac{3197}{10000} \implies \frac{D_2(G)}{n^2} \leqslant \frac{1}{25}. \tag{1}$$

That reference states its results for sufficiently large graphs, and the density condition there is $e(G) \geqslant 0.3197\binom{n}{2}$. To obtain (1), replace each vertex of G by k nonadjacent twins. The resulting graph $G(k)$ is triangle-free and $D_2(G(k)) = k^2 D_2(G)$. For the nontrivial inequality in this identity, let x_i be the fraction of class i on one side of a cut in $G(k)$. Its internal edge count divided by k^2 is

$$\sum_{\{i,j\} \in E(G)} \Big(x_i x_j + (1 - x_i)(1 - x_j) \Big).$$

This is affine in each individual coordinate, so its minimum over the cube occurs at an integral point, corresponding to a cut of G. Also $\rho(G(k)) = \rho(G)$, and $\rho(G) \geqslant 0.3197$ implies $e(G(k)) \geqslant 0.3197\binom{nk}{2}$. Take k sufficiently large and divide the cited bounds by k^2.

Our second input is the certificate from [3]:

$$t(C_4, G) \geqslant \rho(G)^3 - \gamma\rho(G), \qquad \gamma = \frac{3}{64} + \frac{1}{50\,000\,000\,000}. \tag{2}$$

Here $t(C_4, G) = \hom(C_4, G)/n^4$, with noninjective maps included. Appendix B describes the independent coefficient check and the precise sampling argument turning it into this inequality. The max-cut theorem (1) is cited, not reproved by our programs.

We will also use the neighbourhood estimate

$$\beta_*(G) \leqslant \frac{\rho}{8} - \frac{t(C_4, G)}{4\rho} \quad (\rho > 0), \tag{3}$$

which appears with a different four-cycle normalization in [2]. For completeness, if a neighbourhood has size at least $n/2$, an independent fractional half has cost zero. The right side of (3) is nonnegative: the edge-counting identity below and $e(N(u), N(v)) \leqslant e(G)$ imply $t(C_4, G) \leqslant \rho^2/2$. Otherwise every neighbourhood has size less than $n/2$. Fix an edge uv, set $A = N(u)$, $C = N(v)$ and $B = V(G) \setminus (A \cup C)$, and put

$$p = \frac{1/2 - |A|/n}{1 - |A|/n - |C|/n}.$$

The sets A, C are disjoint and independent, and $p \in [0, 1]$. The fractional halves $\mathbf{1}_A + p\mathbf{1}_B$ and $\mathbf{1}_C + (1-p)\mathbf{1}_B$ have costs Q_u, Q_v satisfying

$$(1-p)Q_u + pQ_v = p(1-p)\left(\frac{\rho}{2} - \frac{e(A, C)}{n^2}\right) \leqslant \frac{1}{4}\left(\frac{\rho}{2} - \frac{e(A, C)}{n^2}\right).$$

Average over all edges, using $\sum_{uv \in E(G)} e(N(u), N(v)) = \hom(C_4, G)/2$, to obtain (3) whenever the zero-cost case does not apply.

3 Keeping the degree in the perturbation

Let H be a triangle-free graph with positive vertex weights w_x summing to one. Define

$$\mu = \sum_{xy \in E(H)} w_x w_y, \quad d_x = \sum_{y \in N(x)} w_y, \quad e_x = \sum_{y \in N(x)} w_y d_y, \quad S_x = d_x(e_x - \mu d_x).$$

Assume $\mu > 0$. Let σ denote the total weight of nonisolated vertices. Cauchy–Schwarz in $N(x)$ gives

$$\frac{d_x^2}{e_x} \leqslant \sum_{y \in N(x)} \frac{w_y}{d_y}.$$

All the denominators are positive. Summing with weights w_x and reversing the order of summation proves

$$\sum_{x : d_x > 0} w_x \frac{d_x^2}{e_x} \leqslant \sigma, \qquad \sum_x w_x d_x = 2\mu. \tag{4}$$

For $0 < a \leqslant 1/2$ and $0 < d < 1$, let

$$r(a,d) = \min\left\{\frac{1-a}{1-d}, \frac{a}{d}\right\} \min\left\{\frac{a}{1-d}, \frac{1-a}{d}\right\}. \tag{5}$$

Lemma 2. *Fix $0 < a \leqslant 1/2$ and $K > 0$. Suppose that, for every $d \in (0,1)$ such that $19d > 18\mu$,*

$$K \leqslant r(a,d)\mu d^2 \frac{d + 18\mu}{19d - 18\mu}. \tag{6}$$

Then some nonisolated vertex x satisfies $r(a,d_x)S_x \geqslant K$. Consequently, for every linear functional ℓ, the constant profile a on H can be replaced by a profile $\pi \in [0,1]^{V(H)}$ of the same weighted mean such that

$$\mathcal{Q}(\pi) + \ell(\pi) \leqslant \mathcal{Q}(a) + \ell(a) - K, \qquad \mathcal{Q}(\pi) = \sum_{xy \in E(H)} w_x w_y \pi_x \pi_y.$$

The same assertion holds with constant profile $1-a$.

Proof. Suppose $r(a,d_x)S_x < K$ for all nonisolated x. Then

$$\frac{d_x^2}{e_x} > \frac{d_x^3}{\mu d_x^2 + K/r(a,d_x)}.$$

For $19d_x \leqslant 18\mu$, the right side is positive and hence at least $19d_x/(20\mu) - 9/10$. In the other case, this same lower bound follows by rearranging (6), with positive denominators. Summing the resulting strict inequalities yields

$$\sum_{x:d_x>0} w_x \frac{d_x^2}{e_x} > \frac{19}{10} - \frac{9}{10}\sigma \geqslant \sigma,$$

contradicting (4).

Choose the vertex just obtained and abbreviate $d = d_x$, $\varphi = \mathbf{1}_{N(x)} - d$. Its weighted mean is zero. As $N(x)$ is independent, expansion of the quadratic form gives

$$\mathcal{Q}(\varphi) = -d(e_x - \mu d) = -S_x.$$

The feasible interval for $a + t\varphi$ is $-t_- \leqslant t \leqslant t_+$, where

$$t_+ = \min\{(1-a)/(1-d), a/d\}, \quad t_- = \min\{a/(1-d), (1-a)/d\}.$$

Give its two endpoints probabilities $t_-/(t_+ + t_-)$ and $t_+/(t_+ + t_-)$. Then $\mathbb{E}t = 0$ and $\mathbb{E}t^2 = t_+t_- = r(a,d)$. All linear terms cancel, including ℓ, and the expected change in cost is $-r(a,d)S_x \leqslant -K$. At least one endpoint has that improvement. Replacing a by $1-a$ exchanges t_+ and t_-, leaving their product unchanged. $\square$

The coefficient $19/20$ is a chosen tangent parameter. More generally one can compare the reciprocal expression with $\lambda d/\mu + 1 - 2\lambda$. Our choice allows the verification to use only rational functions; it is not claimed to be optimal.

4 Extending a sparse set

The following argument adapts the greedy extension and neighbourhood sampling in Section 4.5 of [2]. Its starting set need not be independent. For $1/3 \leqslant z \leqslant 1/2$ put

$$J(z) = \begin{cases} 9/128 - z/8, & z \leqslant 3/8, \\ z(1/2 - z)/2, & z \geqslant 3/8. \end{cases}$$

For $1/3 \leqslant A \leqslant 3/8$ define

$$K_A(z) = \begin{cases} (12A - 32z^2 - 12z + 9)/192, & z \leqslant 3/8, \\ (A - 8z^2 + 3z)/16, & z \geqslant 3/8, \end{cases}$$
$$\Phi_A(z) = \max\{J(z), K_A(z)\}.$$

Both J and K_A are continuous and decreasing on this interval.

Lemma 3. *Let G be triangle-free. Suppose its normalized independence number is at most A, where $A \in [1/3, 3/8]$ is rational. If $Z \subseteq V(G)$ has size zn with $1/3 \leqslant z \leqslant 1/2$ and spans εn^2 edges, then*

$$\beta_*(G) \leqslant \varepsilon + \Phi_A(z).$$

Proof. Use a balanced blow-up to assume that $n/8$ and An are integers; all normalized parameters, including the independence number, are unchanged. For the latter claim, an independent set in the blow-up meets only classes whose original vertices are pairwise nonadjacent; taking all vertices of those classes cannot decrease its size. Starting at $B = Z$, repeatedly add a vertex outside B with at most $n/8$ neighbours in B, stopping on reaching $n/2$ vertices or when none is available. Write $b = |B|/n$ and $t = 1/8$. Throughout,

$$\frac{e(B)}{n^2} \leqslant \varepsilon + t(b - z).$$

If $b = 1/2$, this is at most $\varepsilon + J(z)$. Suppose instead the algorithm stops early. Let C be the vertices outside B with more than $bn/2$ neighbours in B. Any two vertices of C have a common neighbour in B, so C is independent and $|C| \leqslant An$. Choose D disjoint from $B \cup C$ with normalized size $D_0 = 1 - A - b$. Every $w \in D$ has normalized degree into B in $(t, b/2]$. Such a choice exists because $|C|/n \leqslant A$. Put $r = 1/2 - b$; in particular, $D_0 - r = 1/2 - A \geqslant 1/8$.

If some $v \in B$ has at least rn neighbours in D, choose rn of them as E. This is an independent set, and $B \cup E$ has normalized cost at most

$$\varepsilon + t(b - z) + \frac{b}{2}(1/2 - b) \leqslant \varepsilon + J(z).$$

The last quadratic has its maximum at $b = 3/8$ when $z \leqslant 3/8$, and at $b = z$ otherwise.

It remains to consider the case that every $v \in B$ has fewer than rn neighbours in D. Choose v uniformly in B, let $E = N(v) \cap D$, $F = D \setminus E$, and $s_v = |E|/n \leqslant r$. Take the fractional half equal to 1 on $B \cup E$ and to $(r - s_v)/(D_0 - s_v)$ on F. The latter ratio lies in $[0, 1]$, since $D_0 > r$. Bounding $e(B, F)/n^2$ by $b|F|/(2n)$, those between E, F trivially, and the edges inside F by Mantel's bound gives its cost at most

$$\frac{e(B) + e(B, E)}{n^2} + \frac{1}{4}(r - s_v)(1/2 + b + 3s_v).$$

Let $s = \mathbb{E}s_v = e(B, D)/(bn^2)$, so $s \leqslant \min\{r, D_0/2\}$. For $u \in D$, write $h_u = |N(u) \cap B|/n \in [t, b/2]$. The inequality $h_u^2 \leqslant (t + b/2)h_u - tb/2$ and double counting give

$$\mathbb{E}\frac{e(B, E)}{n^2} = \frac{1}{bn}\sum_{u \in D} h_u^2 \leqslant (t + b/2)s - \frac{tD_0}{2}.$$

The other term is concave in s_v, so Jensen's inequality bounds the expected cost by ε plus

$$t(b - z) + \frac{r(1 - r)}{4} + (r/2 + t)s - \frac{3}{4}s^2 - \frac{tD_0}{2}. \tag{7}$$

For $b \leqslant 3/8$, the unconstrained maximum in s is at $s = (r + 1/4)/3$, which is no greater than either r or $D_0/2$ because $b \leqslant 3/8 \leqslant 3/2 - 3A$. For $b \geqslant 3/8$, one has $r \leqslant D_0/2$ since $b \geqslant A$, and the maximum on $0 \leqslant s \leqslant \min\{r, D_0/2\}$ is at $s = r$. After substitution, the two expressions are

$$\frac{12A - 32b^2 + 12b - 24z + 9}{192}, \qquad \frac{A - 8b^2 + 5b - 2z}{16},$$

respectively. They coincide at $b = 3/8$. Their derivatives in b are $-(16b - 3)/48$ and $-(16b - 5)/16$, negative on their relevant intervals with $b \geqslant z \geqslant 1/3$. Thus the maximum occurs at $b = z$ and equals $K_A(z)$. One of the sampled halves is no more expensive than its expectation. This proves the lemma. □

Lemma 4. *If a triangle-free graph G has an independent set of normalized size at least $9/25$, then $\beta_*(G) \leqslant 81/3200 < 2543/100000$.*

Proof. Let α be its normalized independence number. For $9/25 \leqslant \alpha \leqslant 3/8$, apply Lemma 3 to a maximum independent set with $A = z = \alpha$ and $\varepsilon = 0$. Here $K_\alpha(\alpha) = 3/64 - \alpha^2/6$, and

$$J(\alpha) - K_\alpha(\alpha) = (\alpha - 3/8)^2/6 \geqslant 0.$$

Hence $\beta_*(G) \leqslant J(\alpha) \leqslant J(9/25) = 81/3200$. For $\alpha \geqslant 3/8$, Theorem 3.6 of [2] gives $\beta_*(G) \leqslant \alpha(1/2 - \alpha)/2 \leqslant 3/128$ when $\alpha \leqslant 1/2$; the case $\alpha \geqslant 1/2$ is immediate. □

5 Neighbourhoods anchored across a maximum cut

Choose a cut X, Y realizing $D_2(G)$, with $|X| \leqslant |Y|$, and retain $c = n/(2|Y|)$, $a = 1 - c$, $\mu = e(Y)/|Y|^2$. Suppose the independence number is at most An, where $A = 9/25$. Define

$$h = \frac{|X|}{|Y|} = 2c - 1, \quad T = 2Ac, \quad k = \min\{T, h\}, \quad Z = \frac{e(X, Y)}{|Y|^2}.$$

We use this construction only when $a \leqslant 1/4$, so $h > 0$. For $u \in Y$, put $d_u = |N(u) \cap Y|/|Y|$ and $\psi_u = |N(u) \cap X|/|Y|$. Optimality of the cut under a single vertex flip implies $\psi_u \geqslant d_u$. Since every neighbourhood in a triangle-free graph is independent, the total degree is at most An. Consequently

$$0 \leqslant d_u \leqslant T/2, \quad d_u \leqslant \psi_u \leqslant k, \quad \mathbb{E}_Y d_u = 2\mu, \quad \mathbb{E}_Y \psi_u = Z.$$

From $(\psi_u - d_u)(k - \psi_u) \geqslant 0$ and Cauchy–Schwarz,

$$\mathbb{E}_Y(d_u\psi_u) \geqslant \mathbb{E}_Y \frac{kd_u^2}{k - \psi_u + d_u} \geqslant \frac{4k\mu^2}{k - Z + 2\mu}. \tag{8}$$

Zero terms with a zero denominator are interpreted by continuity, or omitted before applying Cauchy–Schwarz. The last denominator is positive when $\mu > 0$, since $Z \leqslant k$.

For $x \in X$, let $I_x = N(x) \cap Y$, an independent set in $G[Y]$, and put

$$D_x = \frac{|I_x|}{|Y|}, \quad E_x = \frac{1}{|Y|} \sum_{u \in I_x} d_u, \quad S_x = D_x(E_x - \mu D_x).$$

Double counting gives

$$\mathbb{E}_X D_x = Z/h, \qquad \mathbb{E}_X E_x = \mathbb{E}_Y(d_u\psi_u)/h.$$

Moreover, $0 \leqslant D_x \leqslant T$ and $0 \leqslant E_x \leqslant \min\{\mu, TD_x/2\}$. The first bound on E_x uses the independence of I_x: each edge incident to I_x is counted once.

Lemma 5. *With the maximum-cut notation and independence cap above, assume $0 < a \leqslant 1/4$, $\mu > 0$, $\mu \leqslant T^2/2$, and set*

$$v = \mu(1 - T) - 2\mu^2/T, \qquad W = \frac{4Tk\mu^2}{h(k - Z + 2\mu)} + \frac{vZ}{h} - T\mu + 2\mu^2.$$

For a constant profile a or $1 - a$ on Y, with any added linear functional, a perturbation within Y saves at least $L(a)\max\{W, 0\}/(4c^2)$ in ambient normalization, where $L(a) = \min\{4a^2, a(1 - a)\}$. If $v \geqslant 0$, this conclusion remains true after replacing Z in W by any nonnegative lower bound for Z.

Proof. For $0 \leqslant D \leqslant T$ and $0 \leqslant E \leqslant \min\{\mu, TD/2\}$,

$$D(E - \mu D) \geqslant TE + vD - T\mu + 2\mu^2. \tag{9}$$

Indeed, the difference has coefficient $D - T \leqslant 0$ in E, so it suffices to check its upper endpoint. Put $D_* = 2\mu/T \leqslant T$. For $D \leqslant D_*$ the difference equals $(T - D)(T/2 - \mu)(D_* - D)$; for $D \geqslant D_*$ it equals $\mu(T - D)(D - D_*)$. Both are nonnegative, since $T \leqslant 1$ and $\mu \leqslant T^2/2$ imply $\mu \leqslant T/2$. Average (9) and apply (8) to obtain $\mathbb{E}_X S_x \geqslant W$. Thus some anchor has $S_x \geqslant W$.

The perturbation $\mathbf{1}_{I_x} - D_x$ has mean zero and quadratic cost $-S_x$, exactly as in Lemma 2. The product of its two feasible radii is $r(a, D_x) \geqslant L(a)$ when $0 < D_x < 1$. If $W > 0$, the selected anchor has $D_x > 0$; also $D_x \leqslant T \leqslant 18/25 < 1$. To see the radius inequality, use the three ranges $D_x \leqslant a$, $a \leqslant D_x \leqslant 1-a$, and $D_x \geqslant 1-a$: their products are respectively $a(1-a)/(1-D_x)^2$, $a^2/(D_x(1-D_x))$, and $a(1-a)/D_x^2$. If $W > 0$, choose the better endpoint; if $W \leqslant 0$, keep the original profile. Finally divide the saving by $4c^2$. For $v \geqslant 0$, both Z-dependent terms in W are increasing in Z on $0 \leqslant Z \leqslant k$, proving the last assertion. For a substituted lower bound $Z_0 \leqslant Z$, one has $\mathbb{E}_X S_x \geqslant W(Z) \geqslant W(Z_0)$, so the same anchor-selection argument applies to $\max\{W(Z_0), 0\}$. □

6 The cut construction and the continuous check

Take a cut $V(G) = X \sqcup Y$ realizing $D_2(G)$, with $|X| \leqslant |Y|$ and $I = (e(X)+e(Y))/n^2 \leqslant I_0$. Write

$$c = 1 - a = \frac{n}{2|Y|}, \quad 0 \leqslant a \leqslant \frac{1}{2}, \qquad \mu = \frac{e(Y)}{|Y|^2} \leqslant 4I_0c^2.$$

The profiles $\mathbf{1}_X + a\mathbf{1}_Y$ and $c\mathbf{1}_Y$ are fractional halves. Their costs are respectively

$$B_1 = \frac{a\rho}{2} + cI - \frac{1+a}{4c}\mu, \qquad B_2 = \frac{\mu}{4}. \tag{10}$$

Inside Y, a change to either profile contributes a quadratic cost in $G[Y]$ and an arbitrary linear functional coming from edges to X. The conversion from the normalization on Y to that on G multiplies costs by $|Y|^2/n^2 = 1/(4c^2)$.

By Lemma 4 it remains to treat graphs with independence number less than $(9/25)n$. If $a \leqslant 1/4$, then $z = |X|/n = 1 - 1/(2c) \geqslant 1/3$, and Lemma 3 gives the additional bound

$$\beta_*(G) \leqslant E(a, \mu) := I_0 - \frac{\mu}{4c^2} + \Phi_{9/25}\left(1 - \frac{1}{2c}\right). \tag{11}$$

For $a > 1/4$, regard this candidate as unavailable. When a lower bound $\rho \geqslant R_-$ is also specified, use $Z_0 = 4c^2(R_-/2 - I_0) \leqslant Z$ in Lemma 5, provided its hypotheses and $Z_0 \geqslant 0$, $v \geqslant 0$ hold. With $B(a, \mu)$ defined below, this supplies the auxiliary candidate

$$O(a, \mu) = B(a, \mu) - \frac{L(a)}{4c^2} \max\{W(Z_0), 0\}.$$

An inapplicable auxiliary candidate is simply omitted. The lower bound for Z follows from the exact identity $Z = 4c^2(\rho/2 - I)$.

The following scalar check combines the candidates.

Lemma 6. *Let* $q = 2543/100000$ *and take* (R_-, R, I_0) *from the density table in Appendix A. Define*

$$P(a) = I_0 + a(R/2 - I_0), \qquad B(a, \mu) = \min\left\{P(a) - \frac{1+a}{4(1-a)}\mu, \frac{\mu}{4}\right\}.$$

Whenever $0 < a \leqslant 1/2$, $0 < \mu \leqslant 4I_0(1-a)^2$, $0 < d < 1$, *and* $19d > 18\mu$, *either an available auxiliary candidate is at most* q, *or one has*

$$B(a,\mu) - \frac{r(a,d)\mu d^2(d+18\mu)}{4(1-a)^2(19d-18\mu)} \leqslant q. \tag{12}$$

Proof. Appendix A gives rational upper bounds for the left side on rectangular boxes and a finite subdivision covering the entire domain. The accompanying checker verifies all leaves and the coverage of the subdivision. All acceptance comparisons use rational arithmetic. □

Proposition 7. *If* G *is triangle-free and* $\rho(G) \leqslant 7/20$, *then* $\beta_*(G) \leqslant q$.

Proof. Apply Lemma 4 first. Otherwise the independence cap required in (11) holds. Use the row of the density table containing ρ; the cuts exist by (1). For $a = 0$, the two parts are equal and one has cost at most $I_0/2 < q$. For $\mu = 0$, the second baseline is zero. Thus assume both are positive. Replacing ρ, I in B_1 by R, I_0 can only increase it. If $B(a,\mu) \leqslant q$, a baseline suffices; if an available auxiliary candidate is at most q, its construction suffices. Otherwise put $K = 4c^2(B(a,\mu) - q) > 0$. Lemma 6 supplies (6). Apply Lemma 2 within Y to the better baseline. The resulting fractional half has cost at most $B(a,\mu) - K/(4c^2) = q$. □

7 The remaining densities

For $\rho \geqslant 7/20$, either a zero-cost half exists or (2) and (3) give

$$\beta_*(G) \leqslant \frac{\rho}{8} - \frac{\rho^2}{4} + \frac{\gamma}{4} \leqslant \frac{159}{6400} + \frac{1}{200\,000\,000\,000} < q.$$

The second inequality follows because the quadratic is decreasing for $\rho \geqslant 7/20$. Combining this with Proposition 7 and the rounding argument proves Theorem 1.

A An exact certificate for the continuous inequality

The five rows below cover all densities from zero through 7/20. In the first and last rows the opposite-side candidate is not used. The final column gives total tree nodes, not sampled parameter points.

R_-	R	I_0	Nodes
0	3/10	2/47	5,049
3/10	31/100	2/47	445
31/100	63/200	2/47	1,273
63/200	3197/10000	2/47	1,331
3197/10000	7/20	1/25	467

The starting box is

$$[0, 1/2] \times [0, 4I_0] \times [0, 1]$$

in the coordinates (a, μ, d). Let a subbox have endpoints $[a_-, a_+]$, $[m_-, m_+]$, $[d_-, d_+]$, and put $c_- = 1 - a_+$, $c_+ = 1 - a_-$. A box is discarded if $m_- > 4I_0c_+^2$, or if $19d_+ \leqslant 18m_-$. In boxes with $a_+ \leqslant 1/4$, the extension candidate has the upper bound

$$U_E = I_0 - \frac{m_-}{4c_+^2} + \Phi_{9/25}\left(1 - \frac{1}{2c_-}\right).$$

This uses the fact that $\Phi_{9/25}$ is decreasing in its argument. Such a box is certified when $U_E \leqslant q$. Otherwise the baseline is at most

$$U_B = \min\left\{\frac{m_+}{4},\ I_0 + a_+(R/2 - I_0) - \frac{(1 + a_-)m_-}{4c_+}\right\}.$$

If $U_B \leqslant q$, the whole box is certified. For the opposite-side candidate, set $A = 9/25$ and

$$T_\pm = 2Ac_\pm, \quad h_\pm = 2c_\pm - 1, \quad k_\pm = \min(T_\pm, h_\pm), \quad Z_\pm = 4c_\pm^2(R_-/2 - I_0).$$

Use this candidate only if the row enables it and

$$a_+ \leqslant 1/4, \quad R_- \geqslant 2I_0, \quad m_- > 0, \quad m_+ \leqslant T_-^2/2, \quad Z_+ \leqslant k_-,$$

and

$$v_- = m_-(1 - T_+) - 2m_+^2/T_- \geqslant 0.$$

These conditions justify replacing Z by Z_0 and ensure that all compared denominators are positive. A lower bound for $W(Z_0)$ is

$$W_- = \frac{4T_-k_-m_-^2}{h_+(k_+ - Z_- + 2m_+)} + \frac{v_-Z_-}{h_+} - m_+T_+ + 2m_-^2.$$

Thus a box is also certified when

$$U_B - \frac{L(a_-)}{4c_+^2}\max(W_-, 0) \leqslant q.$$

This comparison does not depend on the degree coordinate d. Here L is increasing on $[0, 1/4]$: both $4a^2$ and $a(1 - a)$ are increasing on that interval.

In the remaining boxes define

$$\mathcal{R}_- = \min\left\{\frac{c_-}{1 - d_-}, \frac{a_-}{d_+}\right\}\min\left\{\frac{a_-}{1 - d_-}, \frac{c_-}{d_+}\right\}.$$

Every factor is a lower bound for its corresponding endpoint radius. The denominator $19d_+ - 18m_-$ is positive in the remaining boxes. An upper bound for the left side of (12) is therefore

$$U_B - \frac{\mathcal{R}_-m_-d_-^2(d_- + 18m_-)}{4c_+^2(19d_+ - 18m_-)}. \tag{13}$$

Endpoint zeros give a valid zero lower bound for the saving. The domains of interest have $0 < d < 1$; no surviving nondegenerate box has $d_- = 1$ or $d_+ = 0$.

The certificate is a preorder binary tree. Symbols $0, 1, 2$ bisect the corresponding coordinate at its rational midpoint; L denotes a leaf. Both children are checked and their union equals their parent, including the shared boundary. The checker rejects any missing nodes, trailing nodes, invalid symbols, or leaves failing all the stated tests. Thus the certificate covers a continuum, including all boundaries, rather than a discrete sampling of points.

Certificate statistic	Combined count
Total tree nodes	8,565
Leaves certified by the baseline	327
Leaves certified by sparse-set extension	841
Leaves certified by opposite-side anchors	736
Leaves with nonpositive tangent	27
Leaves certified by the degree tangent	2,036
Leaves outside the mass constraint	318
Maximum tree depth	28

The verification command below prints the certificate file names. In the order of the density table, their hashes are

```
8a5aa7beda22b8133ce48b0556cac51ee77558cbb4dc370f9a5dba9f17d934ec
f4b4819fc870a0b5e715b7d0c77544091aa6131cc1bca6af4b2680778b70f855
2d755294dbbbdfac6c92a555001870caa25c393ba651fe89d40f795779fe28a6
b8dbe23d14b0c3ceb260cae9f4da0772c1804004c0db31d27e6eb2ac2670a5eb
f8ad90fdb9d0934d91f6f45831c71de7bddc97ae8a309fdd4a177a4fb91c1ee6
```

Replay the stored certificates with Python 3.11:

```
python -S src/verify_paper.py
```

These programs need only the Python standard library. A floating-point heuristic is used only to choose which coordinate to split when generating the tree; it has no role in accepting a leaf. The checker does not use that heuristic. This is a reproducible computer-assisted check, not a formal proof-assistant verification of the Python interpreter or the reduction. A second implementation, `src/audit_intervals.py`, imports no other project modules. It uses generic rational interval operations and recursive tree traversal and checks the same 8,565 nodes independently of the first implementation. It can be run separately:

```
python -S src/audit_intervals.py
```

Its report records exact minimum margins for each leaf type in every row.

The optional script `src/audit_algebra.py` uses SymPy 1.14.0 to expand 12 polynomial identities in the proof exactly; these checks are not numerical samples and are not required by the standard-library replay.

B Independent verification of the four-cycle certificate

The numerical certificate belongs to Amir Sarid [3], at commit

`ba85f319ebb33ec2194ef4c2929a12916329e1e7`

It is retained as `certificates/sarid-clebsch-tangent.json`, with SHA-256

`606b26d835d5b7fe70d077230f8091bbd9677283049a1d0b46a3e3ad30a200ce`

Our separate implementation enumerates all $2^{15} = 32768$ labelled six-vertex simple graphs. Exactly 5,789 are triangle-free. Removing their full relabelling orbits gives 38 representatives, without relying on an external catalogue. The program rederives the coordinate orderings for the certificate's flag vectors and checks all 34 positive rational rank-one terms against all 38 hosts. Every coefficient slack is nonnegative; they are recorded in `results/four-cycle-independent.json`.

Here is why this finite check implies (2) for every finite G. Sample six vertices of G independently with replacement and let H be the graph of adjacencies on their six positions. It is a triangle-free simple graph, even when sampled vertices coincide. The expected indicators of a cycle on positions $1, 2, 3, 4$, of the three edges $12, 34, 56$, and of the edge 12 are respectively $t(C_4, G)$, ρ^3, and ρ. Averaging over all 6! permutations of the positions gives the target coefficient associated with each host.

Each certificate term is an averaged square. For rootless terms, two independent triples realize a product of identical flag functions. For two-root terms, condition on the ordered root and its adjacency; the two disjoint pairs of extension positions are independent, so the conditional expectation of the product is a square. For four-root terms, canonically order the root by its adjacency mask and condition on it. The two leaf positions are again independent and identically distributed. Canonicalization depends only on the root, not on either leaf. Multiplication by the indicator of the specified root type preserves nonnegativity. This reasoning is valid with replacement and does not require the original graph to tend to a graphon limit.

All weights in the certificate are positive. Consequently its expected sum of squares is nonnegative. Coefficientwise domination on every triangle-free host implies that the target expectation is also nonnegative, which is exactly (2).

The independent checker can be run as follows:

```
python -S src/verify_four_cycle.py
```

It uses only the standard library and does not import the original verifier, NetworkX, or a table of unlabelled graphs. Passing this check supports the cited four-cycle inequality; it does not validate unrelated claims in the source repository.

C Limitations and next questions

The bound in Theorem 1 exceeds the conjectured value by $0.00543n^2$. Neither a proof of the conjecture nor a counterexample is provided. The certificates establish the numerical ingredients of the written argument; they do not replace an independent review of its mathematical interpretation. The selection of density ranges and the parameter 19/20 may permit further optimization, but merely optimizing these constants has not been shown to reach 1/50.

Computational materials and AI use

The manuscript, source and computational materials are available at `https://github.com/Komeiji-Shiki/erdos-128-sparse-halves`. File paths refer to that repository or the unpacked review archive.

AI assistance in Codex was used for literature searches, mathematical exploration and derivation, programming, drafting, and the subsequent audit. Earlier methods and Sarid's numerical certificate are attributed where used. The author reviewed the mathematical arguments, the applicability of the cited results, and the computational checks.

References

[1] J. Balogh, F. C. Clemen and B. Lidický, *Max Cuts in Triangle-free Graphs*, extended abstract, 2021. arXiv:2103.14179v1.

[2] A. A. Razborov, *More about sparse halves in triangle-free graphs*, Sbornik: Mathematics **213** (2022), no. 1, 109–128. doi:10.1070/SM9615. Preprint: arXiv:2104.09406v2.

[3] A. Sarid, *Sparse halves in triangle-free graphs: a bound of* 131/5000, preprint dated 4 August 2026, with computational files. Source repository. Accessed 5 September 2026; version pinned in Appendix B.

[4] T. Bloom, *Erdős Problem #128*. `https://www.erdosproblems.com/128`. Accessed 5 September 2026.